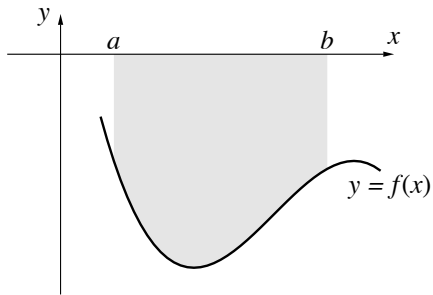


$$\sum_{k=1}^n f(x_k) \Delta x_k = -(\text{shaded area})$$



$$\int_a^b f(x) dx = -(\text{shaded area})$$